



Review

Underwater image enhancement with zero-point symmetry prior and reciprocal mapping[☆]Fei Li^a, Chang Liu^a, Xiaomao Li^{a,b,*}^a School of Mechatronic Engineering and Automation, Shanghai University, Shanghai, 200444, China^b Research Institute of USV Engineering, Shanghai University, Shanghai, 200444, China

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ABSTRACT

Images captured underwater typically exhibit color distortion, low brightness, and pseudo-haze due to light absorption and scattering. These degradations limit underwater image display and analysis, and still challenge the performance of current methods. To overcome these drawbacks, we propose a targeted and systematic method. Specifically, based on a key observation and extensive statistical analysis, we develop a Zero-Point Symmetry Prior (ZPSP): the histograms of channels a and b in the Lab color space, for color-balanced images, exhibit a symmetry distribution around the zero-point. Guided by the ZPSP, a Color Histogram Symmetry (CHS) method is proposed to balance color differences between channels a and b by ensuring they adhere to ZPSP. For channel L, a Reciprocal Mapping (RM) method is proposed to remove pseudo-haze and improve brightness, by aligning its reflectance and illumination components with the Dark Channel Prior (DCP) and Bright Channel Prior (BCP), respectively. Relatedly, it employs a divide-and-conquer strategy, distinguishing underwater image degradations in decomposed sub-images and tackling them individually. Notably, the above-proposed methods are integrated into a systematic enhancement framework, while focusing on targeted optimization for each type of degradation. Benefiting from the proposed strategy and methods, various degradations are individually optimized and mutually promoted, consistently producing visually pleasing results. Comprehensive experiments demonstrate that the proposed method exhibits remarkable performance on various underwater image datasets and applications, also showing good generalization ability. The code is available at: <https://github.com/CN-lifei/ZSRM>.

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1. Introduction

High-quality underwater images play a crucial role in various underwater tasks, such as marine resource exploration, underwater archaeology, and marine biology research [1]. However, light propagation underwater inevitably suffers from scattering and selective absorption, leading to color distortion, low brightness, and pseudo-haze in underwater images [2]. Unfortunately, these degradations typically coexist in underwater images, substantially diminishing their visual quality and negatively affecting a range of underwater tasks. It is hence essential to develop underwater image enhancement techniques to improve the quality of underwater images.

To address these degradations, current methods primarily adopt a multi-stage strategy, wherein degradations are optimized in stages, as illustrated in Fig. 1. Relatedly, image enhancement-based methods [3,4] directly adjust image pixel values to sequentially enhance specific image features, such as color, contrast, and brightness. Image restoration-based methods [5,6] typically remove pseudo-haze using techniques deriving from the daytime dehazing method [7], and further enhance color and brightness by incorporating white balance methods before or after dehazing. Deep learning-based methods [8,9] often employ two-branch networks to compensate for distorted color and reduced contrast, respectively. However, when optimizing for a specific degradation in a single stage, these methods are sensitive to adverse effects from other degradations. Particularly, image restoration-based methods face challenges in achieving similar dehazing performance in underwater images as they do in daytime images, primarily due to the influence of color distortion and non-uniform brightness. Thus, this multi-stage strategy fails to fully exploit the performance of each stage.

Concerning the color correction stage, most methods balance colors by adjusting the histograms or the means of the red, green, and blue channels to be similar, essentially derived from the Grayscale World Hypothesis (GWH) [10]. However, since the GWH [10] ignores wavelength-dependent light absorption, and there is strong coupling among the channels of the RGB color space, such methods potentially introduce undesirable color artifacts, *i.e.*, yellowish or reddish tint, as shown in Fig. 2. (More results are visualized in Figs. 8–12.) Considering the minimal correlation between channels in the Lab color space, a Color Channel Compensation (3C) [11] method is proposed to correct color by pulling the pixels located in channels a and b, which control color information, back to the zero-point representing neutral color. Regrettably, clustering all pixels toward the zero-point reduces color richness. Thus, achieving a balance between enhancing color richness and overcoming color artifacts remains a challenge.

Based on the above observations and analyses, we propose a systematic and targeted method to address underwater image degradations. Specifically, to enhance color richness while minimizing color artifacts, we develop a Zero-Point Symmetry Prior (ZPSP) in the Lab color space.

The ZPSP reveals that for color-balanced images, the histograms of channels a and b typically exhibit symmetric distributions around the zero-point. To validate the reliability of this prior, we conducted the statistics on a total of 2000 color-balanced images across three datasets. The results demonstrate that these color-balanced images indeed highly adhere to this prior. Guided by ZPSP, a Color Histogram Symmetry (CHS) method is proposed to balance color differences by ensuring that underwater images adhere to ZPSP. CHS utilizes the channel independence of the Lab color space to effectively reduce color artifacts and significantly improve color richness by balancing the histogram distribution on both sides of the zero-point. On the other hand, we derive that the Image Formation Model (IFM) exhibits a Bidirectional Mapping Property (BMP), which can be utilized for both image dehazing and brightness enhancement. Correspondingly, built on the BMP of IFM, a Reciprocal Mapping (RM) method is proposed, which removes pseudo-haze and enhances brightness by aligning the reflectance and illumination components on the L channel with the Dark Channel Prior (DCP) [7] and Bright Channel Prior (BCP) [12], respectively.

Notably, as shown in Fig. 1, we employ a divide-and-conquer strategy. Different from the multi-stage strategy, the divide-and-conquer strategy decomposes the underwater image into multiple sub-images to individually represent different degradations, thereby mitigating any adverse effects arising from their interaction. Accordingly, CHS and RM can perform color balancing, dehazing, and brightness enhancement to the maximum performance for the corresponding degraded sub-images. As shown in Fig. 3, we illustrate the results enhanced by our method for several underwater images sampled from different underwater scenes. Without fine-tuning any parameters, our method consistently produces visually pleasing results, benefiting from our strategy and methods. We emphasize the contributions of this work as follows.

- We propose a ZPSP based on a key observation and extensive statistical analysis of color-balanced images. Guided by the ZPSP, a CHS method is further proposed to balance image colors. As shown in Fig. 2, compared to current methods, CHS minimizes color artifacts while balancing and enriching image colors.
- We propose a RM method to enhance image contrast and brightness. Built upon IFM, we achieve effective and robust dehazing performance underwater, comparable to that in daytime, while also generalizing its equally outstanding brightness enhancement capability.
- We propose a divide-and-conquer strategy to distinguish underwater image degradations on different sub-images. By systematically integrating the above-proposed methods into an enhancement framework, we effectively address distinct underwater image degradations individually while achieving mutual promotion.

2. Background

This section describes the basic principle of IFM and reviews the related work.

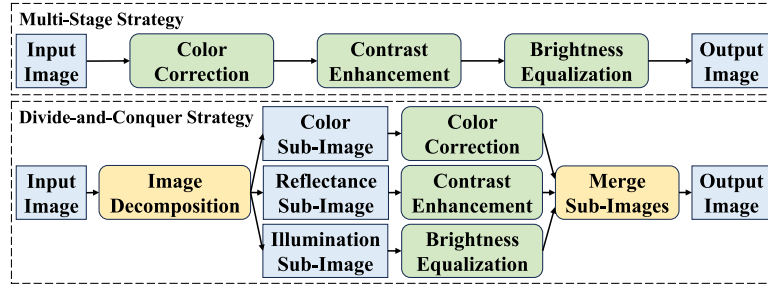


Fig. 1. Illustration of multi-stage and divide-and-conquer strategies. The multi-stage strategy sequentially addresses various underwater image degradations in a concatenated manner. The divide-and-conquer strategy decomposes the input image into sub-images representing different degradations, and then optimizes them individually in parallel. Compared to the multi-stage strategy, the divide-and-conquer strategy can effectively avoid mutual influence among degradations, thus maximizing the optimization of each degradation.

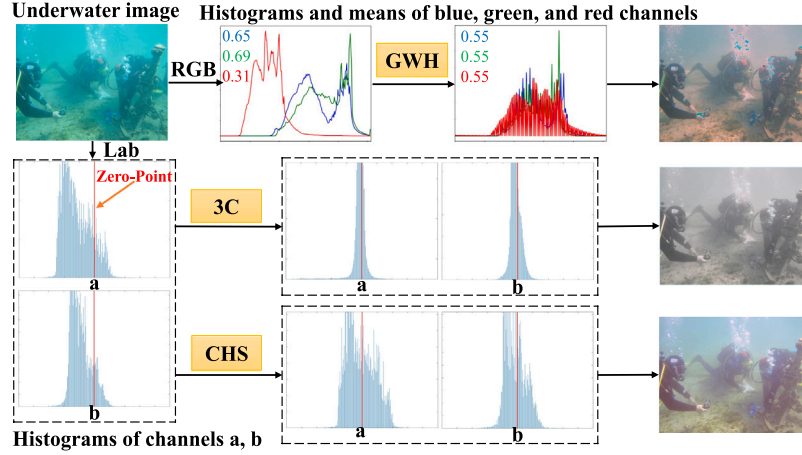


Fig. 2. Illustration of GWH [10], 3C [11], and CHS for balancing color. Technically, GWH [10] balances color by adjusting the histograms or means of red, green, and blue channels in the RGB color space to be consistent. Both 3C [11] and CHS balance color in the Lab color space. The difference is that 3C [11] pulls all pixels of channels a and b back to the zero-point, whereas CHS refines the histograms of channels a and b, aiming to make them symmetrical around the zero-point. From the results, GWH [10] introduces reddish artifacts, 3C [11] generates image with low color richness, while CHS overcomes these issues, producing image with a natural color. (For ease of observation, we use a red vertical line to indicate the zero-point position of channels a and b.)



Fig. 3. Our method enhanced results (bottom row) for several raw underwater images (top row) from the UIEB [13], U45 [14], and UCCS [15] datasets.

2.1. Image formation model

According to the IFM [16], haze imaging can be counted as two components: scene radiance and Backscattered Light (BL), which is formulated as:

$$I(x) = J(x)t(x) + B(1 - t(x)), \quad (1)$$

where $I(x)$ and $J(x)$ are the underwater image and scene radiance that is desired to be restored at the pixel point x , respectively, B is the global BL, and $t(x) \in (0, 1)$ is the Transmission Map (TM), which describes the portion of the scene radiance that is not scattered or absorbed and reaches the camera. The core of restoration-based methods is to recover $J(x)$ based on evaluating BL and TM from $I(x)$.

2.2. Related work

2.2.1. Enhancement-based methods

Enhancement-based methods aim to enhance underwater images by directly adjusting image pixel values without considering the IFM. These methods enhance underwater images mainly from the perspectives of Retinex, histogram equalization, and image fusion. For instance, Zhuang et al. [4] balanced the color differences by aligning mean values of the red, green, and blue channels, then proposed the hyper-laplacian reflectance priors inspired Retinex variational model to enhance underwater image details. Zhang et al. [17] proposed a locally adaptive contrast enhancement method, adjusting input image contrast after locally balancing its color according to a minimum color loss principle. Zhang et al. [18] proposed a principal component analysis

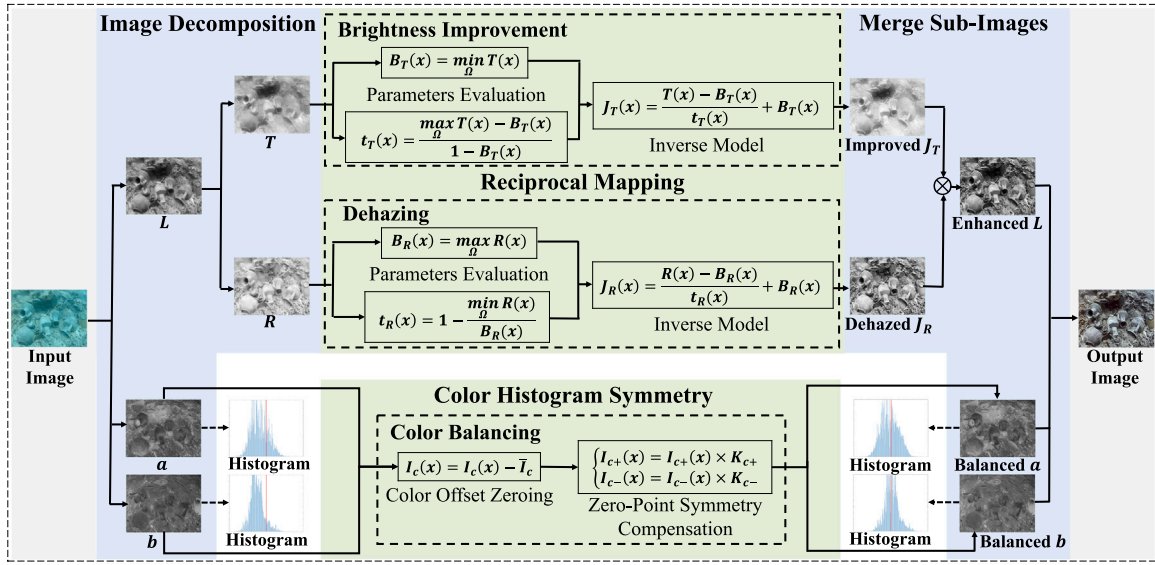


Fig. 4. Flowchart of the proposed method: It employs a divide-and-conquer strategy. In the image decomposition module, the input image is decomposed into four sub-images, with channels a and b representing colors, and components R and T representing scene reflectance and illumination, respectively. The Color Histogram Symmetry (CHS) module balances color differences by adjusting histograms of channels a and b to be symmetric around the zero-point. The Reciprocal Mapping (RM) module separately dehazes R and improves illuminance on T . Finally, the four enhanced sub-images are merged as the final output image.

fusion method to reconstruct a high-visibility underwater image, by integrating the advantages of contrast-enhanced and dehazed sub-images. While these methods exhibit significant performance, they may risk over-enhancement by neglecting the proportional relationship between degradation and object-camera distance.

2.2.2. Restoration-based methods

Restoration-based methods aim to restore high-quality underwater images by inverting the ill-posed IFM. To overcome the ill-posed problem, this line of methods relies on hand-crafted priors to estimate the required model parameters from the degraded input. Notably, these methods are intrinsically inspired by the DCP [7] widely used for outdoor image dehazing. For example, Song et al. [5] proposed a New Underwater Dark Channel Prior (NUDCP) method to remove pseudo-haze, and then introduced a white balance as post-processing to correct color. Liang et al. [19] designed a robust DCP for multiple degradation types of images. Other similar priors include underwater light attenuation prior [20], haze lines prior [21], and color line prior [22]. However, these methods can yield satisfactory results for specific scenes, but they are less effective when these priors do not hold, especially for severe color casts and non-uniform illumination scenes.

2.2.3. Deep learning-based methods

Leveraging the power of neural networks and abundant data, deep learning-based methods excel in converting blurry images into clear ones by learning the feature disparities between low and high-contrast images. These methods are mainly guided by Convolutional Neural Networks (CNNs) and Generative Adversarial Networks (GANs). Cong et al. [23] proposed a physical model-guided GAN that includes two sub-networks, which are used sequentially to obtain color-corrected images and further enhance the authenticity and visual aesthetics of the results. Jiang et al. [8] proposed a deep network, named FA⁺Net, employing a two-stage enhancement structure to enhance colors and restore details of underwater images separately. Liu et al. [24] proposed a dual-supervised GAN that improved image brightness and corrected image colors during the weakly supervised learning stage, while refining image texture details in the intensely directed learning stage. Generally, the effectiveness and robustness of deep-learning-based methods are limited due to the lack of high-quality and sufficient underwater images as training data.

3. Method

This paper presents a divide-and-conquer strategy to address underwater image degradations, with its flowchart depicted in Fig. 4. Initially, the input image is converted from RGB color space to Lab color space, which offers greater stability in handling color and brightness variations. According to the Retinex theory, channel L is further decomposed into Reflectance (R) and Illumination (T) components, which respectively characterize pseudo-haze and non-uniform brightness issues. On one hand, built on the ZPSP, CHS is proposed to balance the color differences of channels a and b . On the other hand, built on the BMP of the IFM, RM is proposed to dehaze R and improve T , thereby enhancing details while maintaining uniform brightness. Finally, the enhanced image is converted from Lab color space to RGB color space as the final result. In what follows, we detail our methodology.

3.1. Zero-point symmetry prior

The proposed prior is inspired by the observation that for color-balanced images, the histogram distribution of channels a and b decomposed from the Lab color space, tends to be symmetric about the coordinate origin (i.e., zero-point). Conversely, images commonly exhibit color deviation when the histograms of channels a and b display an asymmetric distribution along the axes bordering the zero-point. In this paper, we generalize this observation as ZPSP.

To illustrate this observation, we visualize the histograms of channels a and b from multiple samples of color-balanced and underwater images. As shown in Fig. 5, for color-balanced images, the histograms of channels a and b demonstrate symmetric distributions around the zero-point, while the histograms of underwater images deviate from the zero-point.

To validate the reliability of the proposed ZPSP in real-world scenarios, we gathered a total of 2000 images: 890 from the UIEB [13], 500 from LOL [25], and 510 from Nature Footage websites, all representing color-balanced scenes. Building upon previous research, 3C [11] assumes that the mean values of each opponent color channel a and b within a reasonable spatial support closely approximate zero. Therefore, we further assess the reliability of ZPSP by examining whether the histogram distribution of channels a and b is symmetric around the mean value. In our study, we employ skewness as a metric to

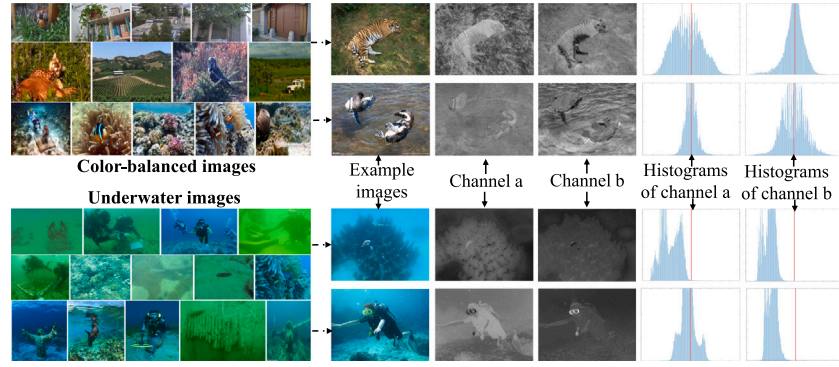


Fig. 5. Some samples of the histogram distribution of channels a and b from color-balanced images and underwater images, respectively.

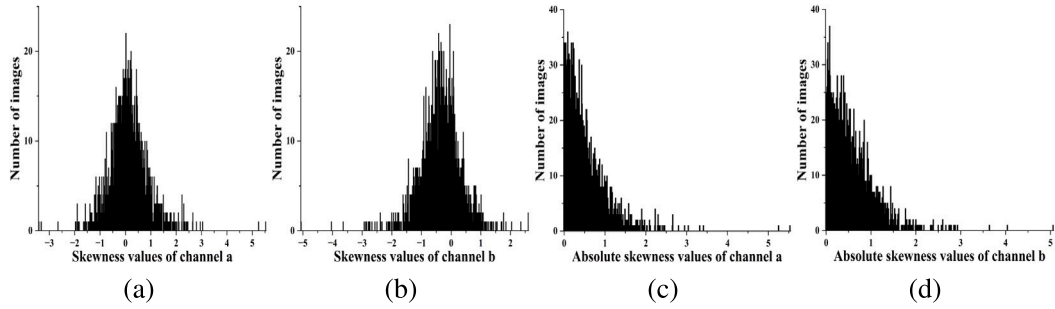


Fig. 6. Statistics of the skewness values from 2000 color-balanced images. (a) and (c) represent the skewness values of channel a and their corresponding absolute skewness values, while (b) and (d) represent the skewness values of channel b and their corresponding absolute skewness values, respectively.

quantify the symmetry of the histogram distribution of channels a and b. Skewness, a concept from probability theory and statistics, measures the asymmetry of the probability distribution of a real-valued random variable about its mean [26,27]. The skewness value can be positive, zero, or negative, with a smaller absolute value indicating closer proximity to symmetry. Specifically, an absolute skewness value of 0 denotes complete symmetrical, with the mean value serving as the axis of symmetry. It should be further pointed out that the absolute skewness value ranges between 0 and 1, signifying the data follows a symmetric distribution, where the absolute skewness between 0 and 0.5 implies that the data is almost completely symmetrical. If the absolute skewness value exceeds 1, it suggests data deviates from symmetry.

The statistics regarding the skewness values of channels a and b across all 2000 color-balanced images are presented in Fig. 6(a) and (b), respectively. The absolute skewness values are depicted in Fig. 6(c) and (d). Fig. 6(a) and (b) illustrate that, for these 2000 color-balanced images, the majority of skewness values fall within the range of -1 to 1 . This indicates that for color-balanced images, the histogram distribution of channels a and b is highly symmetrical about the zero-point. To be more rigorous, from Fig. 6(c) and (d), the absolute skewness values of the channel a range predominantly between 0 – 0.5 , accounting for 61% with values between 0 and 1 constituting 87%. Similarly, for channel b, these values are 55% and 85%, respectively. The statistics strongly provide support for the ZPSP. Again, it is evident that the histograms of the majority of channels a and b exhibit symmetrical distribution centered around the zero-point, indicating that only a small portion of images deviate from our prior.

3.2. Color histogram symmetry

Guided by ZPSP, we propose a CHS method to balance color differences of underwater images. CHS adjusts the histograms of channels a and b to adhere to ZPSP as closely as possible, and it comprises two fundamental steps: (1) color offset zeroing; and (2) zero-point symmetry compensation.

3.2.1. Color offset zeroing

In accordance with ZPSP, it is posited that in assuming that the mean values of each opponent color channel a and b in color-balanced images approximate zero. To this end, we first restore the mean values of channels a and b in underwater images to zero. Given that the direction and magnitude of color shifts are respectively indicated by the sign and value of the mean values, we define the color offset of channels a and b as their mean values:

$$\bar{I}_c = \frac{1}{HW} \sum_{i=1}^H \sum_{j=1}^W I_c(i, j), c \in (a, b), \quad (2)$$

where H and W are the height and width of channels a and b, respectively. Mathematically, color offset zeroing is achieved by subtracting the mean values of channels a and b:

$$I_c(x) = I_c(x) - \bar{I}_c. \quad (3)$$

It should be emphasized that this step only shifts the histograms of channels a and b to both sides of the zero-point, ensuring their mean values become null, without changing the shape of the histograms.

3.2.2. Zero-point symmetry compensation

To ensure that the histograms of channels a and b fit symmetrically, we propose a zero-point symmetry compensation method inspired by the GWH [10] to finely adjust the histograms on both sides of the zero-point. Specifically, we calculate the average values of the positive and negative axes on both sides of the zero-point of channels a and b, denoting them as $\bar{I}_{c+}$ and $\bar{I}_{c-}$, respectively. Subsequently, by obtaining the means of the positive and negative axes, we derive the correction factor K as follows:

$$K_c = \frac{|\bar{I}_{c+}| + |\bar{I}_{c-}|}{2}, \quad (4)$$

where $||$ represents the absolute value symbol. The gain coefficients of the positive and negative axes are as follows:

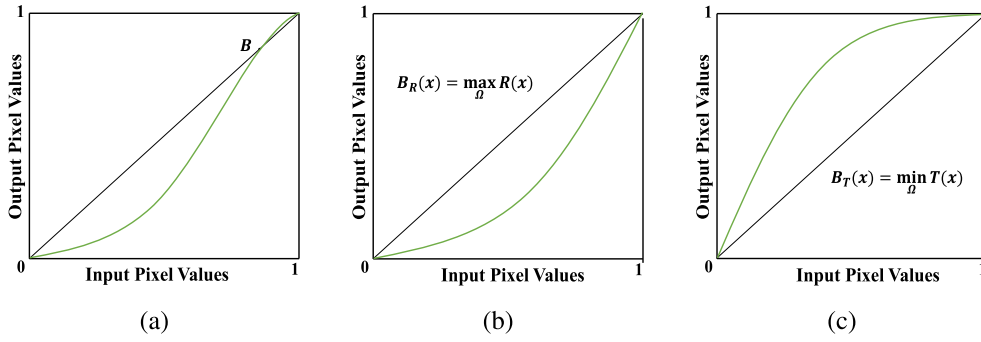


Fig. 7. Examples of different mapping functions from input to output images based on the IFM. Input pixel values (as depicted by the black diagonal) are mapped to output pixel values (as depicted by the green curve) according to the BMP.

$$\begin{cases} K_{c+} = \frac{K_c}{|I_{c+}|}, \\ K_{c-} = \frac{K_c}{|I_{c-}|}. \end{cases} \quad (5)$$

Subsequently, we apply the following linear gain to the positive and negative axes of channels a and b based on their gain coefficients:

$$\begin{cases} I_{c+}(x) = I_{c+}(x) \times K_{c+}, \\ I_{c-}(x) = I_{c-}(x) \times K_{c-}. \end{cases} \quad (6)$$

On the basis of the mean values of channels a and b being zero, this step further fine-tunes the histograms on both sides of the zero-point by calculating the gain coefficients of their positive and negative axes to align with ZPSP.

3.3. Bidirectional mapping property

Eq. (1) can be reformulated as:

$$J(x) = I(x) + \left(\frac{1}{t(x)} - 1\right)(I(x) - B). \quad (7)$$

Since $t(x) \in (0, 1)$, it follows that $(1/t(x) - 1) > 0$. Therefore, when $(I(x) - B) \leq 0$, $J(x) \leq I(x)$, and the opposite situation occurs when $(I(x) - B) > 0$. As a result, the IFM exhibits bidirectional effects: brighter regions ($I(x) \geq B$) become even brighter, while darker regions ($I(x) < B$) become darker. We refer to this phenomenon as the BMP of IFM. Once B is given, the IFM consistently demonstrates this BMP, wherein the input pixel values are mapped to the output pixel values in opposite directions with B as the boundary.

Proof of Monotonicity: Based on Eqs. (1) or (7), we solve the derivative of $J(x)$ with respect to $I(x)$ as:

$$\frac{\partial J(x)}{\partial I(x)} = \frac{1}{t(x)}. \quad (8)$$

Because $t(x) \in (0, 1)$, $1/t(x)$ is always greater than 0. This means that based on the IFM, the mapping curve from input image $I(x)$ to output image $J(x)$ is monotonically increasing. Based on the inherent properties of the HFM, including its monotonic increasing and bidirectional mapping, we illustrate the mapping curve from input image $I(x)$ to output image $J(x)$ as shown in Fig. 7(a).

3.4. Reciprocal mapping

In addition to color balancing for channels a and b, we propose a RM method that includes both upward mapping and downward mapping to separately remove pseudo-haze and improve brightness on channel L. Specifically, according to Retinex theory, we decompose channel L into two components, R and T. In this decomposition, pseudo-haze predominantly manifests in the R component, while low brightness is primarily reflected in the T component. Considering the pseudo-haze resembling daytime haze in R, we propose a downward mapping

method based on DCP [7] to remove the pseudo-haze in R. On the other hand, for the poor illumination affecting T, we propose an upward mapping method guided by BCP [12] to improve the brightness in T. Specific details are as follows:

Dehazing R based on downward mapping method: After mitigating the influence of color distortion and poor illumination on R, we leverage the robust dehazing capability of DCP [7] to remove the pseudo-haze effect on R. According to the haze imaging principle formulated in Eq. (1), similar assumptions can be made about R:

$$R(x) = J_R(x)t_R(x) + B_R(x)(1 - t_R(x)), \quad (9)$$

where, $R(x)$ and $J_R(x)$ represent R and restored R, $B_R(x)$ and $t_R(x)$ are the BL and TM of R, respectively.

For hazy images, DCP [7] typically selects the brightest 0.1% of pixels to define the BL. However, unlike daytime hazy images, underwater images often exhibit non-uniform illumination. In such cases, BL becomes a spatially varying matrix map rather than a global vector. To better apply DCP [7] to underwater images, we introduce the local background light $B(x)$ to replace the consistent background light B. Specifically, the local background light $B(x)$ can be estimated by searching for the brightest pixel within a local patch of the reflectance component R, expressed as:

$$B_R(x) = \max_{x \in \Omega} R(x), \quad (10)$$

where Ω is a local patch centered at x , with a size of 15×15 .

For evaluating the parameter TM, we follow DCP [7]. Specifically, it is assumed that the haze-free image $J_R(x)$ meets the following prior conditions:

$$\min_{x \in \Omega} J_R(x) = 0. \quad (11)$$

Taking the min operation in the local patch on Eq. (9), we have:

$$\min_{x \in \Omega} R(x) = \min_{x \in \Omega} J_R(x)t_R(x) + B_R(x)(1 - t_R(x)). \quad (12)$$

Putting Eq. (11) into Eq. (12), we can estimate the $t_R(x)$ simply by:

$$t_R(x) = 1 - \frac{\min_{x \in \Omega} R(x)}{B_R(x)}. \quad (13)$$

Here, guided filter [28] is used to refine $t_R(x)$. After obtaining the required model parameters $t_R(x)$ and $B_R(x)$, we invert Eq. (9) to recover high-quality scene reflectance $J_R(x)$ by:

$$J_R(x) = \left(\frac{R(x) - B_R(x)}{t_R(x)}\right) + B_R(x). \quad (14)$$

Based on the core idea of DCP [7], we propose a downward mapping method to remove pseudo-haze effects on the R component. Notably, we transform BL from a global vector B to a spatially varying matrix map $B(x)$ to better generalize the dehazing performance of DCP [7] to underwater images. Additionally, since $B_R(x)$ represents the maximum

value of each local patch, any pixel value in the input image $R(x)$ is less than or equal to $B(x)$. Accordingly, as per the BMP of the IFM, when the input pixel value is less than BL, as shown in Fig. 7(b), the model performs a downward mapping when restoring the haze-free image $J_R(x)$. Therefore, we refer to the proposed method as the downward mapping method.

Enhancing T based on upward mapping method: On the other hand, due to the visual similarity between inverted low-light images and hazy images, some works [29–31] borrow the DCP [7] to enhance low-light images by incorporating two intensity-inversion operations. This inspires us to address the issue of image brightness based on the IFM. However, it is worth noting that the IFM exhibits a BMP, allowing direct enhancement of image brightness without the need for image inversion. Therefore, we formulate T based on Eq. (1) as follows:

$$T(x) = J_T(x)t_T(x) + B_T(x)(1 - t_T(x)), \quad (15)$$

where, $T(x)$ and $J_T(x)$ represent T and enhanced T , $B_T(x)$ and $t_T(x)$ are the pseudo-BL and pseudo-TM of T , respectively.

In detail, to enable the IFM to effectively enhance and equalize image brightness, we assign a small value to BL. This encourages the IFM to demonstrate an upward mapping property, as illustrated in Fig. 7(c). Correspondingly,

$$B_T(x) = \min_{x \in \Omega} T(x). \quad (16)$$

Unlike reflectance R , illumination T under optimal exposure conditions adheres to the BCP [12]:

$$\max_{x \in \Omega} J_T(x) = 1. \quad (17)$$

Taking the max operation in the local patch on Eq. (15), we have:

$$\max_{x \in \Omega} T(x) = \max_{x \in \Omega} J_T(x)t_T(x) + B_T(x)(1 - t_T(x)). \quad (18)$$

Putting Eq. (17) into Eq. (18), we can estimate the $t_T(x)$ simply by:

$$t_T(x) = \frac{\max_{x \in \Omega} T(x) - B_T(x)}{1 - B_T(x)}. \quad (19)$$

Here, guided filter [28] is used to refine $t_T(x)$. Further, $J_T(x)$ is solved by inverting Eq. (15).

$$J_T(x) = \left(\frac{T(x) - B_T(x)}{t_T(x)} \right) + B_T(x). \quad (20)$$

Based on BCP [12], we propose an upward mapping method to enhance the brightness of the T component. Since $B_T(x)$ represents the minimum value of each local patch, any pixel value in the input image $T(x)$ is greater than or equal to $B(x)$. Accordingly, as per the BMP of the IFM, when the input pixel value is greater than $B(x)$, as shown in Fig. 7(c), the model performs an upward mapping to improve the brightness of the T component. Therefore, we refer to the proposed method as the upward mapping method. Furthermore, given that the enhancement mapping curves for R and T are reciprocal to each other, we refer to the method as RM.

After dehazing R and enhancing T , they are combined back into channel L through point multiplication. Finally, channels a and b with balanced color distribution, along with the channel L comprising brightness and contrast-optimized illuminance T and reflectance R , are recombined in the Lab color space and converted to the RGB color space as the final result.

4. Experiments

In this section, we first describe the experimental settings, and then conduct extensive experiments on multiple datasets to verify the effectiveness of the proposed method.

4.1. Experimental setup

4.1.1. Benchmark datasets

Our method is tested on the UIEB [13], U45 [14], and UCCS [15] datasets. UIEB [13] includes 890 underwater images of diverse underwater scenes and their corresponding reference images selected from the optimal results of prolific methods. U45 [14] contains 45 underwater images, where color casts, low contrast, and haze-like effects of underwater degradation are all covered. UCCS [15] contains 300 underwater images of bluish, blue-green, greenish tones, and severe blurring. These datasets thoroughly validate the color correction, contrast enhancement, and balance non-uniform illumination performance of various methods, while verifying the effectiveness and robustness of different methods for enhancing underwater images in multiple scenarios.

4.1.2. Comparison methods

Our method is compared with 10 underwater image enhancement methods, including enhancement and restoration methods: NUDCP [5], MLE [17], ICSP [32], WWPF [3], and PCFB [18]; and deep learning-based methods: FUnIEGAN [33], UWnet [34], TACL [35], FA⁺Net [8], and PUGAN [23].

4.1.3. Evaluation metrics

We adopt four commonly-used image quality evaluation metrics for quantitative analysis, including Underwater Image Quality Measurement (UIQM) [36], Colorfulness Contrast Fog density index (CCF) [37], Natural Image Quality Evaluator (NIQE) [38], and Fog Aware Density Evaluator (FADE) [39]. UIQM [36] is a linear composition of underwater image sharpness measure (UISM), underwater image colorfulness measure (UICM), and underwater image contrast measure (UIConM). CCF [37] is a weighted combination of colorfulness, contrast, and fog density indexes. A larger UIQM [36] or CCF [37], the better the color balance, clarity, and contrast of the image. NIQE [38] uses space domain natural scene statistics to measure various degradation. FADE [39] assesses the performance of methods designed to enhance the visibility of degraded images. For NIQE [38] or FADE [39], smaller values represent better image quality.

4.1.4. Experiment categories

To comprehensively evaluate the effectiveness of our method, we carried out six types of experiments, including qualitative, quantitative, and detail enhancement comparisons, ablation analysis, application comparisons, and generalization tests.

4.2. Experiments results and discussion

4.2.1. Qualitative comparisons

In this section, we present qualitative comparisons by visualizing partial enhancement results obtained from our method alongside those from several excellent underwater image enhancement methods. As shown in Figs. 8–11, these results are selected from a variety of underwater scenes, presenting dim light, pseudo-haze, bluish and greenish, color distortion, edge detail loss, and non-uniform brightness under artificial lighting.

Fig. 8 shows the underwater scene characterized by a bluish, hazy appearance, and edge detail loss. It is evident from Fig. 8 that ICSP [32] fails to demonstrate its performance in such an underwater environment. NUDCP [5], FUnIEGAN [33], and UWnet [34] introduce a yellowish tint, resulting in poor contrast enhancement. Although MLE [17], WWPF [3], and PCFB [18] yield favorable enhancement results, they tend to produce over-enhanced and less natural outcomes. Moreover, TACL [35], FA⁺Net [8], and PUGAN [23] demonstrate outstanding color correction capabilities; however, the enhanced images still encounter some degree of blurring, particularly noticeable in the details where this blurriness is more pronounced. In contrast, the

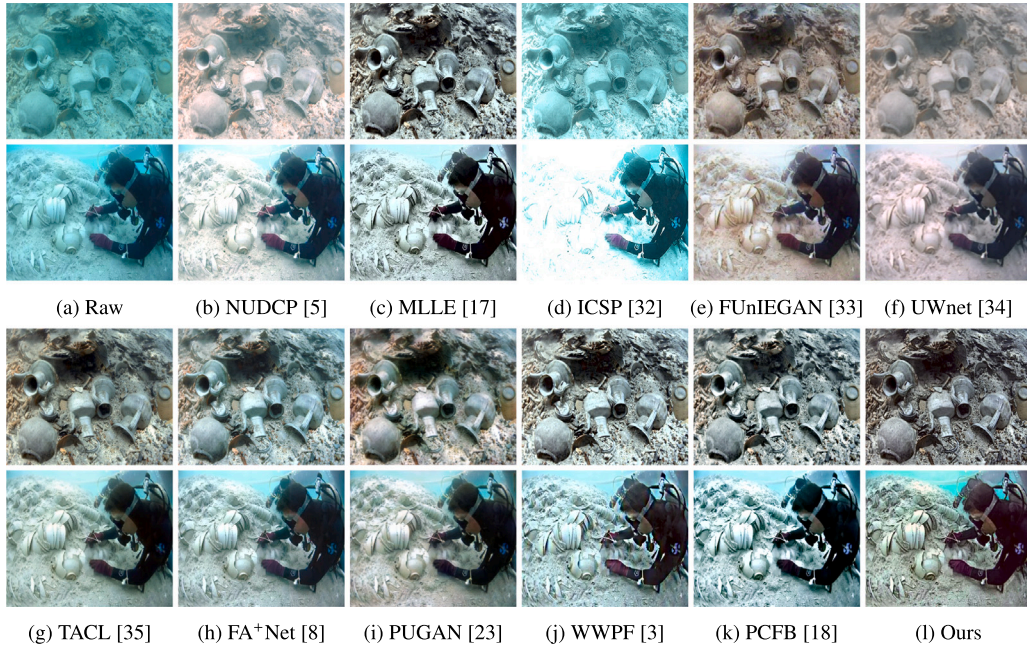


Fig. 8. Qualitative comparisons on the UIEB [13] dataset. From (a) to (l) are (a) raw images, the results produced by (b) NUDCP [5], (c) MLLE [17], (d) ICSP [32], (e) FUnIEGAN [33], (f) UWnet [34], (g) TACL [35], (h) FA⁺Net [8], (i) PUGAN [23], (j) WWPF [3], (k) PCFB [18], and (l) our method.

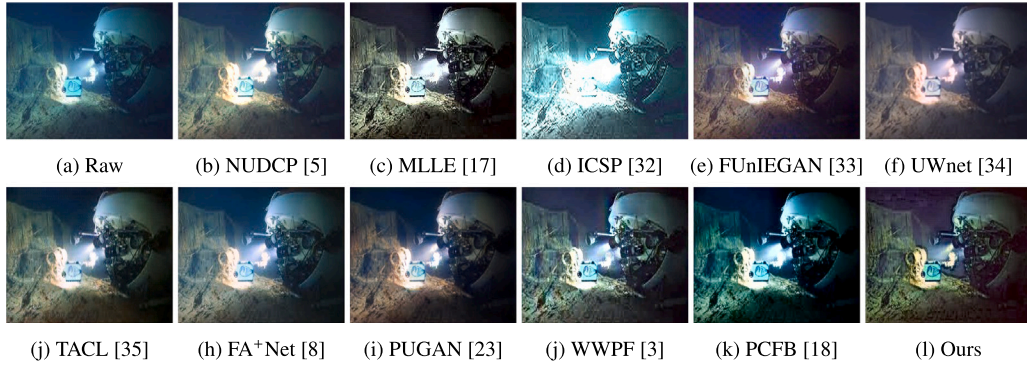


Fig. 9. Qualitative comparisons on the UIEB [13] dataset. From (a) to (l) are (a) raw images, the results produced by (b) NUDCP [5], (c) MLLE [17], (d) ICSP [32], (e) FUnIEGAN [33], (f) UWnet [34], (g) TACL [35], (h) FA⁺Net [8], (i) PUGAN [23], (j) WWPF [3], (k) PCFB [18], and (l) our method.

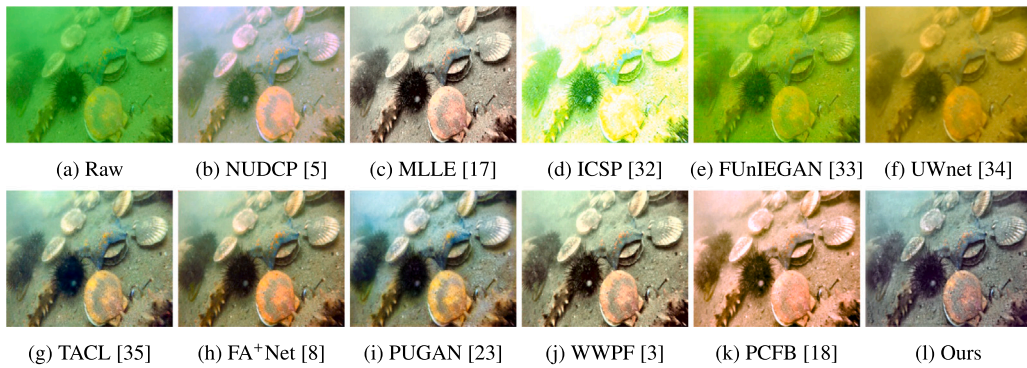


Fig. 10. Qualitative comparisons on the U45 [14] dataset. From (a) to (l) are (a) raw images, the results produced by (b) NUDCP [5], (c) MLLE [17], (d) ICSP [32], (e) FUnIEGAN [33], (f) UWnet [34], (g) TACL [35], (h) FA⁺Net [8], (i) PUGAN [23], (j) WWPF [3], (k) PCFB [18], and (l) our method.

proposed method yields enhancement results with truer colors, more contrast, and an effective highlight of the image's fine textures.

Moreover, Fig. 9 represents an underwater extreme imaging scene with artificial lighting. FUnIEGAN [33], UWnet [34], and PUGAN [23] introduce a yellowish appearance. As for ICSP [32], it introduces

severe over-enhancement in terms of brightness. The enhanced results of TACL [35], NUDCP [5], and FA⁺Net [8] exhibit similarities with increased brightness, but still retain a hazy appearance. PCFB [18] fails to effectively balance colors and significantly reduces brightness in areas away from light sources. While MLLE [17] and WWPF [3]

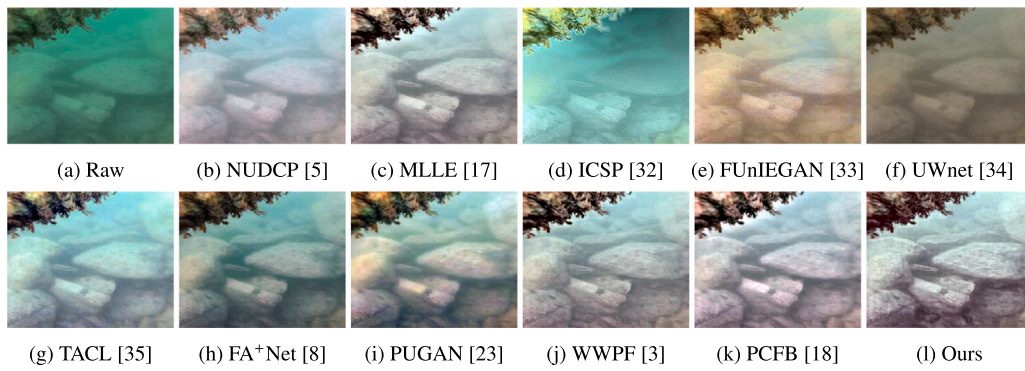


Fig. 11. Qualitative comparisons on the UCCS [15] dataset. From (a) to (l) are (a) raw images, the results produced by (b) NUDCP [5], (c) MLLE [17], (d) ICSP [32], (e) FUnIEGAN [33], (f) UWnet [34], (g) TACL [35], (h) FA+Net [8], (i) PUGAN [23], (j) WWPF [3], (k) PCFB [18], and (l) our method.

perform well in boosting contrast and brightness, it over-enhances the light source in its center, erasing its details. In comparison to these methods, the proposed method allows for efficient restoration of color and contrast in the presence of such extreme scenes. However, our method in this scenario has a limitation: it does not sufficiently enhance the brightness of areas outside the light sources. Considering the performance of various enhancement methods in this scene, this limitation is a common constraint among current underwater image enhancement techniques.

Furthermore, Fig. 10 shows a dim light, haze, and greenish scene. Images enhanced by ICSP [32] and FUnIEGAN [33] still exhibit a greenish hue, while UWnet [34] and PCFB [18] introduce a slightly yellowish and reddish appearance, respectively. While NUDCP [5], TACL [35], FA+Net [8], PUGAN [23], and WWPF [3] effectively correct the color deviation, their performance in contrast enhancement is not noteworthy. In addition, MLLE [17] achieves good results in this scene, where its performance being particularly significant in terms of brightness enhancement. However, the image enhanced by the proposed method exhibits remarkable detail and natural color.

The scene illustrated in Fig. 11 represents poor brightness, bluish, and blurry. Referring to Fig. 11, the contrast cannot be effectively enhanced by NUDCP [5], ICSP [32], FUnIEGAN [33], UWnet [34], and PUGAN [23], while images enhanced by ICSP [32], TACL [35], and FA+Net [8] still show a bluish appearance. MLLE [17], WWPF [3], and PCFB [18] demonstrate a significant contrast enhancement performance, while PCFB [18] introduces a reddish color artifact. In contrast, the proposed method achieves better color restoration, contrast enhancement, and edge detail preservation.

To sum up, it is clear from the results displayed in Figs. 8–11 that ICSP [32] can improve brightness, but falls short in addressing color deviation and blurring issues. NUDCP [5], FUnIEGAN [33], and UWnet [34] offer only modest contrast enhancement and prove ineffective in scenes with pronounced color deviation and blurring. TACL [35], FA+Net [8], and PUGAN [23] exhibit potential in contrast enhancement and eliminate bluish and greenish phenomena, yet their performance requires further improvement for significant impact. MLLE [17], WWPF [3], and PCFB [18] demonstrate strong performance in terms of contrast enhancement, but they introduce over-enhancement in bright areas, with PCFB [18] also introducing reddish color artifacts. Compared to other methods, the proposed method can achieve better results in various underwater scenarios, underscoring its effectiveness.

4.2.2. Quantitative comparisons

Table 1 presents the quantitative results of our method and several comparative methods on UIEB [13], U45 [14], and UCCS [15] benchmarks. In addition, four image evaluation metrics, UIQM [36], CCF [37], NIQE [38], and FADE [39], are used to evaluate the performance of each method. Specifically, UIQM [36] is a comprehensive

evaluation metric designed for underwater images, which combines aspects such as sharpness, color, and contrast. The higher the value of UIQM [36], the better the image quality. According to the data in Table 1, our method performs best in the UIQM [36] metric on the UIEB [13] and UCCS [15] datasets, and ranks second on the U45 [14] dataset. This indicates that our method produces images with higher quality in terms of sharpness, color fidelity, and contrast. CCF [37] focuses on evaluating color information, where our method achieves the best results on the UIEB [13] dataset and ranks third on the U45 [14] and UCCS [15] datasets, second only to MLLE [17] and PCFB [18]. This indicates that compared to current state-of-the-art underwater image enhancement methods, our method also shows stable and highly competitive performance in color correction. NIQE [38] assesses the naturalness of the image, and our method performs best in this indicator, achieving the highest scores on both the UIEB [13] and U45 [14] datasets, indicating that the enhanced images produced by our method have a high degree of naturalness. This benefits from the excellent performance of our method in three aspects: color, contrast, and brightness. FADE [39] focuses on detecting the degree of haze in the image to evaluate visibility, and our method also performs well in this metric, achieving first, second, and third scores on the UCCS [15], U45 [14], and UIEB [13] datasets, respectively. This indicates that our Reciprocal Mapping (RM) algorithm has achieved significant effectiveness in removing the pseudo-haze effect in underwater images. These data demonstrate that our method has the most competitive performance in terms of color correction, contrast enhancement, dehazing, and improving brightness across multiple underwater image datasets.

4.2.3. Detail enhancement comparisons

Restoring blurry details is essential for generating high-quality images, and it best reflects the performance of different methods. Hence, we conduct a detail enhancement comparison experiment, as illustrated in 12.

From a global perspective, compared to the raw underwater image, our results exhibit balanced colors and high contrast, demonstrating the significant performance of the proposed CHS and RM methods. Compared to other state-of-the-art methods, our result overcomes color artifacts such as yellowish or reddish hues seen in ICSP [32], FUnIEGAN [33], UWnet [34], and PCFB [18]. This further demonstrates the reliability of the ZPSP in mitigating color artifacts.

Compared to the input image, overall, our method demonstrates significant improvements in color and contrast; locally, our method significantly enhances the detail information in the enlarged regions within the red and green boxes. Compared to other methods, our method achieves the best results in detail enhancement. It is worth noting that restoring image details presents significant challenges. For example, methods like NUDCP [5], ICSP [32], FUnIEGAN [33], UWnet [34], and PUGAN [23] perform worse than the original underwater image in terms of detail recovery. While WWPF [3], PCFB [18], and MLLE [17]

Table 1

Quantitative comparisons of different methods on the UIEB [13], U45 [14], and UCCS [15] datasets. The best result is in red whereas the second best result is in blue under each case.

Methods	UIEB [13]				U45 [14]				UCCS [15]			
	UIQM↑	CCF↑	NIQE↓	FADE↓	UIQM↑	CCF↑	NIQE↓	FADE↓	UIQM↑	CCF↑	NIQE↓	FADE↓
NUDCP [5]	2.4604	23.9018	3.9918	0.7814	2.9816	22.8224	4.3241	0.5529	2.2832	15.1956	4.7336	1.1369
MLLE [17]	3.7454	48.2163	4.1116	0.3887	4.6003	60.0319	4.8303	0.2402	4.0675	38.3700	4.8576	0.4132
ICSP [32]	1.3764	25.1700	5.6536	0.7303	1.1856	20.2711	7.8273	0.9950	0.5201	17.9373	6.4495	0.5746
FUnIEGAN [33]	4.7209	21.4530	4.7622	0.3612	4.1698	19.4501	4.4482	0.3074	4.0564	17.6932	4.6615	0.4484
UWnet [34]	4.3238	19.3115	6.3885	0.6850	4.0482	19.5476	6.1042	0.6247	3.6342	12.1922	6.3916	1.1955
TACL [35]	3.6449	24.3459	3.6622	0.4248	4.4912	32.0674	4.0453	0.2770	3.7650	25.1726	3.5399	0.4105
FA ⁺ Net [8]	3.3547	23.9123	3.8082	0.5469	4.3735	27.4508	4.0310	0.3659	3.1786	20.4181	4.7341	0.8100
PUGAN [23]	4.3252	27.1958	4.7205	0.3189	4.3898	28.3959	4.6604	0.3141	4.2198	26.5910	5.0491	0.5107
WWPF [3]	3.7780	40.9264	3.8987	0.3711	4.2263	48.3045	3.9471	0.2391	3.9469	34.0076	4.5924	0.4098
PCFB [18]	3.6394	47.2392	3.6252	0.3380	16.6052	56.1060	4.0016	0.2108	3.8827	34.3967	4.5246	0.4929
Ours	5.9911	49.5104	3.5005	0.3583	4.6170	51.8687	3.8941	0.2232	4.2921	34.0331	4.6533	0.4071

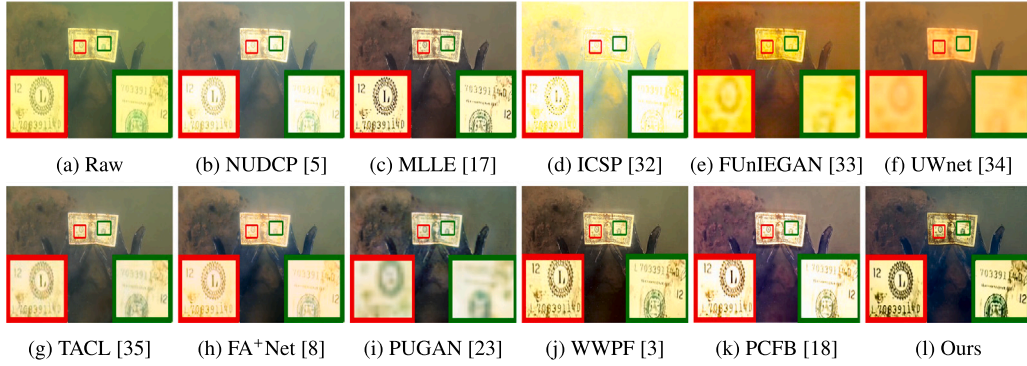


Fig. 12. Visual comparisons of detail enhancement. From (a) to (l) are (a) raw images, the results produced by (b) NUDCP [5], (c) MLLE [17], (d) ICSP [32], (e) FUnIEGAN [33], (f) UWnet [34], (g) TACL [35], (h) FA⁺Net [8], (i) PUGAN [23], (j) WWPF [3], (k) PCFB [18], and (l) our method.

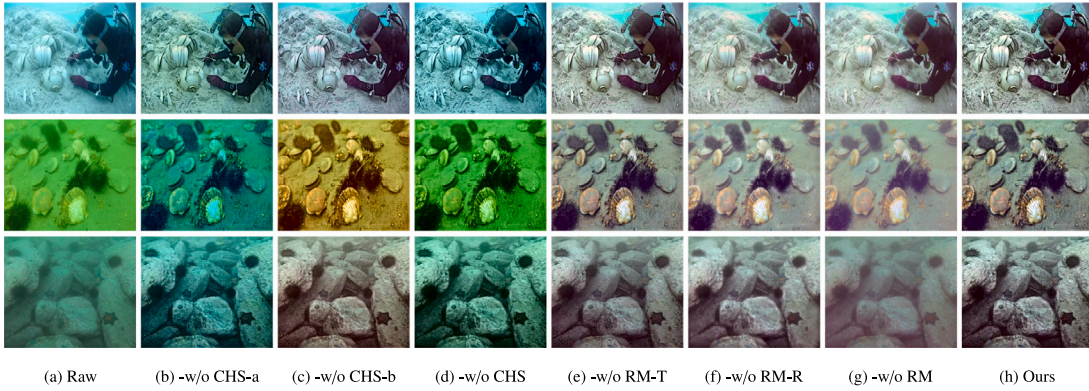


Fig. 13. Qualitative ablation results for each key component of our method on the UIEB [13], U45 [14], and UCCS [15] datasets. The test images from the UIEB [13], U45 [14], and UCCS [15] are in the first column. (a) Raw images. (b) -w/o CHS-a. (c) -w/o CHS-b. (d) -w/o CHS. (e) -w/o RM-T. (f) -w/o RM-R. (g) -w/o RM. (h) Ours (full model).

improve detail enhancement compared to the original underwater image, their results still fall short of ours. This demonstrates that our method enhances both the overall image quality and significantly improves its detailed textures.

4.2.4. Ablation analysis

To demonstrate the effectiveness of each component in our method, we conduct the following ablation study on UIEB [13], U45 [14], and UCCS [15] datasets, including (1) our method without Color Histogram Symmetry of a channel (-w/o CHS-a); (2) our method without Color Histogram Symmetry of b channel (-w/o CHS-b); (3) our method without Color Histogram Symmetry (-w/o CHS); (4) our method without Reciprocal Mapping of Illumination T (-w/o RM-T); (5) our method without Reciprocal Mapping of Reflectance R (-w/o RM-R); (6) our method without Reciprocal Mapping (-w/o RM).

The visualization comparison of our method's ablation study on the UIEB [13], U45 [14], and UCCS [15] datasets is shown in Fig. 13. The following visual results can be clearly observed: (1) -w/o CHS-a fails to correct the greenish tone; (2) -w/o CHS-b introduces a yellowish hue; (3) -w/o CHS enhances contrast but cannot change the color of the image; (4) -w/o RM-T enhances both color and contrast, but the degree of enhancement still needs to be further strengthened; (5) -w/o RM-R effectively removes color deviations but cannot significantly improve contrast; (6) -w/o RM corrects color deviation, but fails to enhance contrast; (7) The full model with all key components achieves the most satisfactory visual results in terms of color, contrast, and brightness. The above results show the efficiency of each module in our method as well as the success of our adopted divide-and-conquer strategy, which fully leverages the strengths of each module.

The quantitative scores of the ablated models on the UIEB [13], U45 [14], and UCCS [15] datasets are presented in Table 2. As shown

Table 2

Quantitative comparisons of different methods on the UIEB [13], U45 [14], and UCCS [15] datasets. The best result is in red whereas the second best result is in blue under each case.

Methods	UIEB [13]				U45 [14]				UCCS [15]			
	UIQM↑	CCF↑	NIQE↓	FADE↓	UIQM↑	CCF↑	NIQE↓	FADE↓	UIQM↑	CCF↑	NIQE↓	FADE↓
-w/o CHS-a	3.1652	42.6284	3.8224	0.3655	4.2506	44.3115	4.3647	0.2902	1.1146	30.9351	4.7575	0.2764
-w/o CHS-b	3.9033	35.5858	3.7898	0.4267	4.4098	45.7615	4.2948	0.2479	4.2106	26.7782	4.7997	0.4861
-w/o CHS	2.9764	44.4555	3.5722	0.2684	4.3382	49.8714	3.9553	0.1621	1.4580	40.2516	4.6812	0.2659
-w/o RM-T	3.7911	38.5510	3.8665	0.5679	4.5474	44.4005	4.4494	0.2821	3.8036	24.3800	4.8301	0.6889
-w/o RM-R	4.0158	32.7804	3.8807	0.7398	4.9330	34.6196	3.8450	0.4612	3.5845	21.4586	4.6156	0.8696
-w/o RM	3.3066	27.6707	4.2514	1.0486	4.1838	25.0569	4.3553	0.9076	2.3709	11.8643	5.1419	1.9192
Ours	5.9911	49.5104	3.5005	0.3583	4.6170	51.8687	3.8941	0.2232	4.2921	34.0331	4.6533	0.4071



Fig. 14. Application examples of local feature points matching. From (a) to (l) are (a) raw images, the results produced by (b) NUDCP [5], (c) MLE [17], (d) ICSP [32], (e) FUnEGAN [33], (f) UWnet [34], (g) TACL [35], (h) FA+Net [8], (i) PUGAN [23], (j) WWPF [3], (k) PCFB [18], and (l) our method.

in Table 2, the full model including all components achieves the best results, indicating that each component helped our method perform better. To be more precise, the best result is from the full model, followed by -w/o CHS. This indicates that compared to CHS, the RM algorithm has a more significant impact on image enhancement, specifically by significantly improving the quality of underwater images through pseudo-haze removal and brightness enhancement. Therefore, images processed with RM achieve scores in various metrics that are second only to those of the full model. Additionally, CHS also plays a crucial role in balancing underwater image colors. With the integration of CHS, the full model achieves optimal performance overall. This indicates that each module in our proposed method contributes positively to improving the quality of underwater images.

4.2.5. Application comparisons

In this section, the proposed method serves as a pre-processing for underwater image keypoint matching, segmentation, keypoint detection, and object detection to further validate its efficacy regarding underwater applications.

Keypoint matching technology consists of connecting two images with similar scenes using the same feature points. The more significant the enhancement performance of the method is, the more feature points the image matches will be. Therefore, the Scale-Invariant Feature Transform (SIFT) [40] is applied to match the underwater images. As depicted in Fig. 14, the image enhanced by the proposed method achieves a significant increase in the number of keypoint matches. Compared to traditional methods, this method is second only to MLE [17]. Compared with deep learning methods, our method achieves the best results. Image segmentation refers to the technology of separating homogeneous areas within an image. YOLOv8 is adopted to segment underwater images. The results, illustrated in Fig. 15, highlight that the underwater images enhanced by the proposed method are more consistent and have richer segmentation results. Image keypoint detection aims to calculate the number of keypoints that characterize image contrast. Moreover, the SIFT [40] is applied to detect these keypoints. As displayed in Fig. 15, the underwater images enhanced by

the proposed method significantly increase the number of keypoints. This demonstrates that our approach can recover local features of underwater images. Furthermore, underwater object detection aims to locate and recognize objects appearing in underwater scenes. The clearer the underwater image is, the higher the detection accuracy of the algorithm will be. Once again, YOLOv8 is utilized to detect underwater images, the objects present in the enhanced underwater image are more accurately detected.

4.2.6. Generalization tests

Since the degradation issues addressed by our proposed methods are commonly present across various scenarios, employing the divide-and-conquer strategy maximizes the performance of each algorithm in addressing individual problems. Therefore, our method achieves better enhancement results for underwater images, while exhibiting good generalization ability for both nighttime and daytime haze images without fine-tuning any parameters.

Fig. 16 presents the results of enhancing daytime hazy images. Compared to the original hazy images, the results generated by our method significantly reduce haze, thereby markedly improving image contrast. While compared to three state-of-the-art daytime image dehazing methods (DCP [7], Tarel [41], and NPBG [42]), our method demonstrates more pronounced performance in enhancing image contrast and brightness, without introducing color artifacts in the sky region. Therefore, our method exhibits good generalization ability in enhancing daytime hazy images.

Similarly, Fig. 17 presents the results of enhancing nighttime hazy images. From the enhanced results, our method exhibits two significant features for the nighttime haze images: firstly, it effectively removes the presence of haze in the images, and secondly, it significantly balances the image colors, thereby improving the visibility and naturalness of nighttime haze images. Compared to three state-of-the-art nighttime image dehazing methods (NIM [43], GMLC [44], and MRP [45]), our approach demonstrates competitive enhancement performance, especially in contrast enhancement, where our method stands out the most.

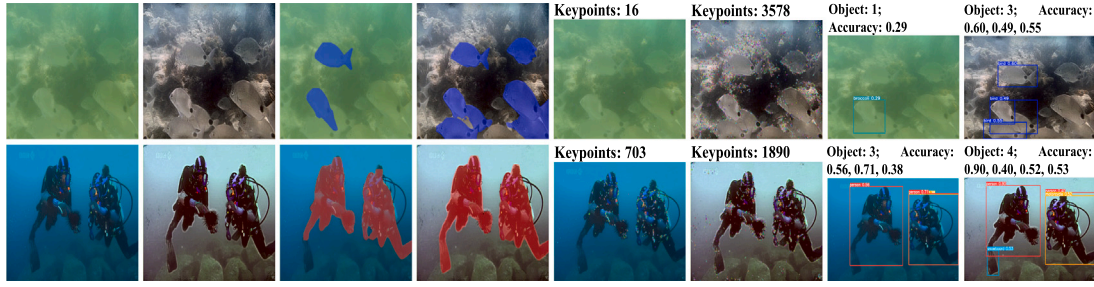


Fig. 15. Application examples of underwater image segmentation, keypoint detection, and object detection. From left to right are the raw underwater images, the images enhanced by our method, and the corresponding application results for the raw input and the enhanced result, respectively.

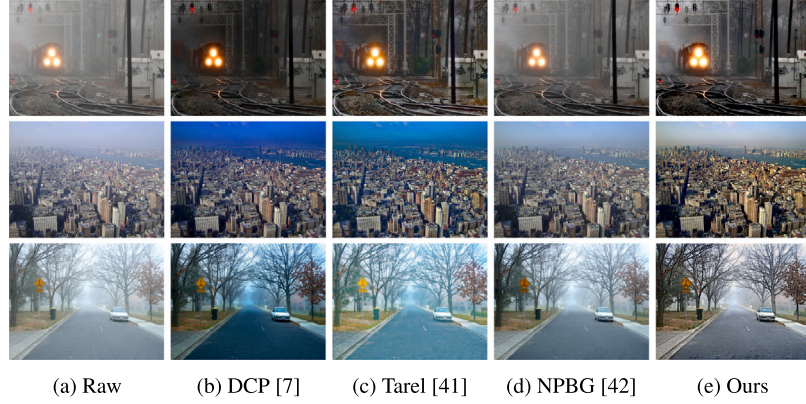


Fig. 16. Generalization test. Comparison of enhancement results of low-light images. (a) Raw daytime haze images. The results are produced by (b) DCP [7], Tarel [41], (d) NPBG [42], and (e) the proposed method.

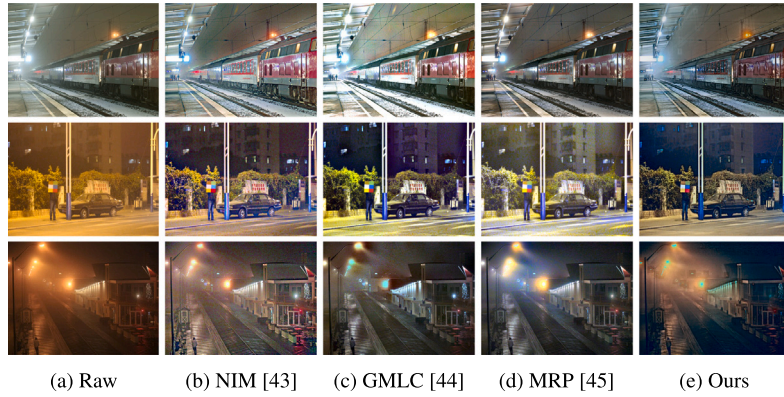


Fig. 17. Generalization tests. Comparison of enhancement results of night haze images. (a) Raw night haze images. The results are produced by (b) NIM [43], (c) GMLC [44], (d) MRP [45], and (e) the proposed method.

5. Conclusion

In this paper, we present a divide-and-conquer strategy to tackle underwater image degradations. This strategy separates different degradations on decomposed sub-images, thereby avoiding their mutual influence and maximizing enhancement for each degradation. To balance the distorted color, we present a ZPSP, a more reasonable and reliable prior for characterizing the property of color-balanced images. CHS is then proposed to apply this prior to underwater images, balancing the color differences of underwater images becomes simpler and more effective. Furthermore, leveraging the BMP of IFM, a RM method is proposed to achieve stable and efficient dehazing and brightness enhancement performance on underwater images. Extensive experiments conducted on various benchmarks demonstrate the effectiveness and generalization of our method.

Limitation and future work. One of our limitations is that for images with active light sources, our method encounters challenges in enhancing brightness in areas distant from the light source. Currently, we are exploring ways to further improve the performance of brightness enhancement by overcoming the insensitivity of our method to these areas. Building upon existing underwater image enhancement strategies and methods, we hope that our work can provide a new perspective and make a contribution for future endeavors in this field.

CRediT authorship contribution statement

Fei Li: Writing – original draft, Writing – review & editing, Conceptualization, Methodology, Formal analysis, Software, Validation, Visualization, Data curation,. **Chang Liu:** Conceptualization, Writing – original draft, Formal analysis, Validation. **Xiaomao Li:** Writing – review & editing, Validation, Methodology, Resources, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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